Suraj Rao

surajrao@umd.edu | +1 2406678312 | Washington, DC | LinkedIn | GitHub | Website | Open to Relocate

EDUCATION

University of Maryland College Park, *United States* *Sep 2025 - May 2027*
Master of Science in Applied Machine Learning [GPA: 3.9/4]

Indian Institute of Information Technology Pune, *India* *Aug 2020 - Aug 2024*
Bachelor of Technology in Computer Science and Engineering [GPA: 8.09/10]

EXPERIENCE

AI Research Assistant | ***UMD MIRAGE Lab,*** *Maryland, United States* *Jan 2026 - Present*

- Built end-to-end lattice defect detection pipeline using YOLO26 on ~21K images across 3 materials for edge deployment
- Addressed 1:9 class imbalance using data augmentation and PyTorch weighted sampling, improving minority class detection
- Improved thickness classification by analyzing failure cases and refining training strategies, reducing false positives by 45%
- Achieved 96.3% mAP across 6 classes and deployed on NVIDIA Jetson using ONNX and TensorRT for real-time inspection

AI Engineer | ***IIT Delhi,*** *Delhi, India* *Oct 2024 - Jul 2025*

- Evaluated transformer-based vision models for occluded vehicle detection in Indian road scenarios
- Curated high-occlusion (>50%) data from RSUD20K, IDD, BDD100K via pseudo-labeling and built a Tkinter GUI for false positive correction, reducing annotation time from 7 min to 15 sec (~96%) [GitHub]
- Fine-tuned 4 SOTA transformer detectors (DETR, DINO, Grounding DINO, Co-DETR) on 4 NVIDIA GPUs using distributed multi-GPU training and benchmarked their performance using mAP and recall

AI Engineer Intern | ***Mantra Softech,*** *Bangalore, India* *Dec 2023 - Oct 2024*

- Built and optimized a video surveillance pipeline by deploying real-time YOLOv8 for people detection and counting
- Reduced model size 4× via quantization, enabling parallel deployment across 4 camera streams using multiprocessing
- Increased throughput from 10 to 35 FPS via frame filtering, batching, and model profiling; deployed on RSC-101 [Product]
- Evaluated DALL-E Mini, Stable Diffusion, CLIP to build a text-to-video retrieval system for ~300+ hours of video footage

PROJECTS

AI-Powered Investor Intelligence Platform | *Azure OpenAI, Azure AI Search, FastAPI, AKS* [GitHub]

- Built a RAG-based platform to extract 8 financial KPIs from company reports with a chatbot for investor queries
- Implemented semantic chunking with Azure OpenAI embeddings and Azure AI Search for context retrieval
- Developed the FastAPI backend with Azure Database for PostgreSQL; deployed the application on AKS using Docker, GitHub Actions, and Azure Container Registry

Multimodal Semantic Search Engine | *CLIP, FAISS, Streamlit, AWS SageMaker* [GitHub]

- Built a multimodal semantic search engine using CLIP for text-to-image and image-to-image retrieval on 5.4K+ images
- Generated CLIP embeddings and indexed them with FAISS, improving search speed by 40% and enabling <200 ms latency
- Developed a Streamlit app with text and image query support, returning top-5 semantically relevant results
- Improved usability using HTML/CSS and deployed a scalable, containerized pipeline on AWS SageMaker

Multi-Objective YOLOv8 Optimization for Edge Deployment | *ONNX, TensorRT, YOLOv8* [GitHub]

- Built an ONNX-based pipeline to export, quantize and benchmark 144 YOLOv8 configs by latency, model size and mAP
- Implemented Pareto-based optimization under deployment constraints; 51 of 144 configurations satisfied the constraints
- Converted the optimal ONNX configuration to a TensorRT engine and deployed it on NVIDIA GPU and Jetson Orin Nano

SKILLS

Programming:	Python, C++, SQL, HTML, CSS
Gen AI:	LLMs, Transformers, RAG, AI Agents, MCP, OpenAI API, Azure OpenAI
Frameworks:	PyTorch, HuggingFace, LangChain, LangGraph, FastAPI, OpenCV, Scikit-learn, NumPy, Pandas
Backend & Cloud:	Docker, Azure AI Search, ChromaDB, FAISS, Kubernetes, AWS SageMaker, CI/CD
Optimization:	CUDA, ONNX, TensorRT, Quantization, Pruning, EdgeAI, vLLM
Tools:	Git, Linux, Streamlit, MLflow, n8n